HACKATHON · DATA, PRIVACY AND TRUST

MEDGUARD AI

Adaptive cybersecurity for healthcare data

Don’t just verify the user. Verify the context.



Data Privacy | Zero Trust | Risk Engine | Human Review

THE 10-SECOND STORY

A doctor logs in. Password correct. Is everything fine?



The attacker doesn’t need to break the hospital if they can borrow a trusted identity.

THE SECURITY GAP

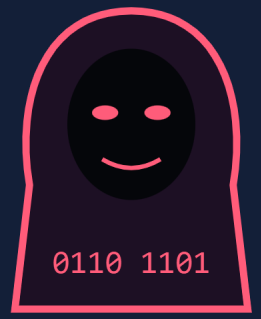
Same password. Very different story.



Real doctor

Known laptop
Hospital network
10:15 AM
5 normal records

Low risk → Allow



Stolen credentials

Unknown device
Unusual location
2:47 AM
Hundreds of records

Risk 91 → Human review

Simulated locally with synthetic data. No real hospital is ever attacked.

CONTEXT-AWARE ACCESS

9 simple questions before access

Identity
Who is asking?

Role
What authority do they have?

Device
Known or unknown?

Location
Expected place or network?

Time
Normal working hours?

Purpose
Why do they need it?

Volume
How many records?

Sensitivity
How private is the data?

Behavior
Matches their normal pattern?

A valid identity is one signal. It is not a permanent trust decision.

RISK ENGINE

Simple rules. A clear score. A fair response.

91

out of 100

HIGH RISK

0–30	**Low risk: Allow** Doctors work without extra friction
31–60	**Medium risk: Verify** Ask for extra proof
61–100	**High risk: Human review** A person decides

Unknown device · Unusual location · Unusual time · Mass access · Sensitive data

ARCHITECTURE

Where MEDGUARD sits in a hospital



Today’s prototype: web interface, simulated context, transparent risk policy and audit workflow.

PRIVACY BY DESIGN

Security must not become a privacy risk



Data minimization

Ask only for needed data



Synthetic data

No real patient records



Least privilege

Only what role and purpose need



Auditability

Decisions leave evidence



Separation

Security data stays clinical-free



Safe defaults

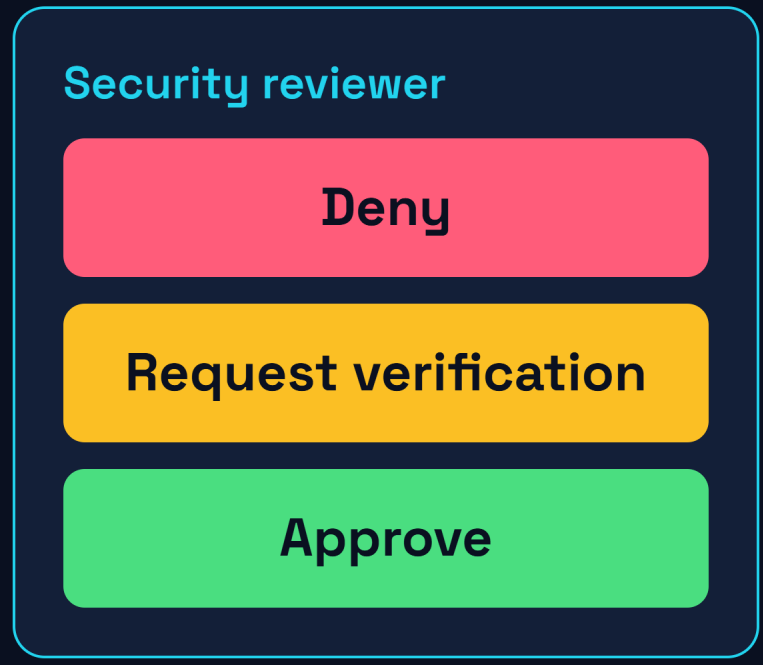
Risky requests get escalated

HUMAN IN THE LOOP

Should an algorithm block everything suspicious?

No.

Healthcare is high stakes. A false positive can stop real care. A false negative can expose patient data. So high risk goes to a trained person.



Human oversight is not a weakness. It is a safety feature.

THREAT COVERAGE

One engine. Six access threats.



Stolen credentials

Valid account, abnormal context



Insider misuse

Authorized user, abnormal volume



Unknown device

New device wants sensitive data



Unusual location

Unexpected network or place



Mass record access

Many records, very fast



Off-hours access

Sensitive access at odd times

LIVE DEMO

Attack to response in five steps

1. **Command Center**
   See the security state
2. **Threat Lab**
   Pick Stolen Credentials
3. **Run simulation**
   Score jumps to 91/100
4. **Human Review**
   Click Deny
5. **Audit Ledger**
   Decision recorded

Valid credentials, abnormal context, accountable decision.

BUILT HONESTLY

What we made and where it goes next

Prototype today

- React and Vite web app
- Transparent rule-based risk engine
- Synthetic patient data, stored locally
- Human review and audit trail
- No real hospital systems touched

NOW — **Prototype**
Works on synthetic data

NEXT — **Pilot**
Identity and EHR/API integration

THEN — **Scale**
Behavior baselines, ML anomalies

LATER — **SOC and SIEM**
Plug into security operations

IMPACT

Who benefits?



Patients

Better protection from unnecessary exposure



Clinicians

Fast access in normal work. Extra checks only when risk rises



Security teams

Clear, ranked, explainable alerts instead of raw noise

Privacy + clinical continuity + security response, together.

MEDGUARD AI

Does this access request make sense right now?

A valid identity should not automatically mean unlimited trust.

Thank you.